Software Engineering - Comprehensive MCQs

SECTION 1: USER STORIES

Basic Concepts

Q1. Who coined the term "user stories" in Extreme Programming?

- A) Robert Martin
- B) Kent Beck
- C) Barry Boehm
- D) Grady Booch

Answer: B) Kent Beck

Q2. A user story should be written from which perspective?

- A) Developer perspective
- B) Tester perspective
- C) Stakeholder/Client/User perspective
- D) Project manager perspective

Answer: C) Stakeholder/Client/User perspective

Q3. Which of the following is NOT a characteristic of user stories?

- A) Simple and not technical
- B) Understandable
- C) Detailed technical specifications
- D) Written in natural language

Answer: C) Detailed technical specifications

Q4. What is the primary purpose of a user story?

- A) To provide detailed technical specifications
- B) To describe functionality valuable to a user
- C) To replace all documentation
- D) To eliminate the need for testing

Answer: B) To describe functionality valuable to a user

Q5. Which template format is used to write user stories?

- A) "As a [system], I want to [action], so I can [benefit]"
- B) "As a [user role], I want to [goal], so I can [reason]"
- C) "As a [developer], I want to [code], so I can [deploy]"
- D) "As a [manager], I want to [control], so I can [monitor]"

Answer: B) "As a [user role], I want to [goal], so I can [reason]"

User Story Components

Q6. What are the three parts of a user story card?

- A) Title, Description, Acceptance Criteria
- B) Description, Conversation, Confirmation
- C) Planning, Design, Implementation
- D) Requirements, Design, Testing

Answer: B) Description, Conversation, Confirmation

Q7. What is the purpose of the "Confirmation" section in a user story card?

- A) To discuss the story with stakeholders
- B) To convey what tests will be carried out to confirm completion
- C) To provide written description of the feature
- D) To assign the story to developers

Answer: B) To convey what tests will be carried out to confirm completion

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Q8. What is the "Conversation" section in a user story card used for?

- A) Storing technical implementation details
- B) Capturing further information and details of conversations
- C) Writing code
- D) Defining the project schedule

Answer: B) Capturing further information and details of conversations

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Q9. Which of the following best describes acceptance criteria?

- A) The number of developers required
- B) Criteria defining how we know when we're done
- C) The project budget
- D) The deployment timeline

Answer: B) Criteria defining how we know when we're done

User Story Detail and Splitting

Q10. How detailed should a user story be?

- A) Extremely detailed with all technical specifications
- B) Detailed enough for the team to start working and establish clarifications at development time
- C) Very brief with minimal information
- D) Detailed only in the code comments

Answer: B) Detailed enough for the team to start working and establish clarifications at development time

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Q11. Detail can be added to user stories in two ways. Which of the following is one of them?

- A) By adding more developers to the team
- B) By splitting a user story into multiple smaller stories
- C) By increasing the project budget
- D) By hiring more testers

Answer: B) By splitting a user story into multiple smaller stories

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Q12. What is the "vertical slicing" approach in user story splitting?

- A) Breaking the story horizontally into layers
- B) Slicing the story vertically so each slice has a bit of all components
- C) Dividing stories by technical complexity
- D) Splitting stories by team members

Answer: B) Slicing the story vertically so each slice has a bit of all components

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Q13. When splitting user stories, what should each slice contain?

- A) Only UI components
- B) Only database components
- C) A bit of all required components (like a vertical slice of cake)
- D) Only business logic

Answer: C) A bit of all required components (like a vertical slice of cake)

Q14. According to INVEST criteria, what does "I" stand for?

- A) Important
- B) Independent
- C) Integrated
- D) Identified

Answer: B) Independent

Q15. According to INVEST criteria, what does "V" stand for?

- A) Verifiable
- B) Valid
- C) Valuable
- D) Viewable

Answer: C) Valuable

Q16. What does the "S" in INVEST represent?

- A) Specific
- B) Small
- C) Simple
- D) Staged

Answer: B) Small

Q17. According to INVEST criteria, what does "T" stand for?

- A) Testable
- B) Technical
- C) Timely
- D) Traced

Answer: A) Testable

Q18. What does "E" in INVEST stand for?

- A) Estimatable
- B) Essential
- C) Evaluated
- D) Enhanced

Answer: A) Estimatable

Q19. According to INVEST, what does "N" stand for?

- A) Necessary
- B) Negotiable
- C) Noted
- D) Named

Answer: B) Negotiable

Q20. What does the final "T" in INVEST stand for?

- A) Technical
- B) Targeted
- C) Testable
- D) Timed

Answer: C) Testable

Q21. What is the key principle behind user story negotiation?

- A) It is a contract with fixed requirements
- B) It is a reminder for features to discuss and clarify
- C) It is a technical specification
- D) It is a legally binding document

Answer: B) It is a reminder for features to discuss and clarify

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Q22. Why should user stories be estimatable?

- A) To provide detailed implementation steps
- B) To establish project budget
- C) To provide enough information to estimate without being too detailed
- D) To determine team size

Answer: C) To provide enough information to estimate without being too detailed

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Example Analysis

Q23. In the example "As a registered user I want to log in, so I can access subscriber-only content," what is the "reason"?

- A) Log in
- B) Access subscriber-only content
- C) Being registered
- D) Using the system

Answer: B) Access subscriber-only content

Q24. In the user story template, the "who" represents:

- A) The system architect
- B) The user role
- C) The project manager
- D) The developer

Answer: B) The user role

Q25. What is the purpose of including the "why" (reason) in a user story?

- A) To confuse developers
- B) To provide clarity, influence feature function, and inspire other useful features
- C) To add unnecessary information
- D) To slow down development

Answer: B) To provide clarity, influence feature function, and inspire other useful features

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SECTION 2: TRADITIONAL SOFTWARE DEVELOPMENT MODELS

Waterfall Model

Q26. The Waterfall model is also known as:

- A) Spiral model
- B) Sequential model
- C) Agile model
- D) Iterative model

Answer: B) Sequential model

Q27. Which characteristic best describes the Waterfall model?

- A) Highly flexible and allows frequent changes
- B) Phases overlap significantly
- C) Each phase must be completed before the next begins
- D) Continuous feedback from users

Answer: C) Each phase must be completed before the next begins

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Q28. A major disadvantage of the Waterfall model is:

- A) Too much iteration
- B) Difficult to detect errors until late stages
- C) Continuous client involvement
- D) Requires minimal documentation

Answer: B) Difficult to detect errors until late stages

Incremental vs. Iterative Development

Q29. In incremental development, what is the first step?

- A) Build the complete system immediately
- B) Design the complete product first, then slice into chunks
- C) Test the system before design
- D) Deploy to production immediately

Answer: B) Design the complete product first, then slice into chunks

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Q30. What is the key characteristic of incremental development?

- A) Complete requirements must be known upfront
- B) Modules are built and integrated separately
- C) Continuous stakeholder feedback is essential
- D) No design phase is needed

Answer: B) Modules are built and integrated separately

Q31. Iterative development differs from incremental development because:

- A) It requires more documentation
- B) It uses a process of successive approximation with feedback
- C) It doesn't require any planning
- D) It eliminates the need for testing

Answer: B) It uses a process of successive approximation with feedback

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Q32. In iterative development, what happens after building a prototype?

- A) The project is immediately deployed
- B) Feedback is collected and incorporated into the next iteration
- C) No further changes are allowed
- D) The documentation is finalized

Answer: B) Feedback is collected and incorporated into the next iteration

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Q33. Which approach is more suitable when requirements are not fully understood?

- A) Waterfall
- B) Incremental only
- C) Iterative
- D) Big Bang

Answer: C) Iterative

Q34. The key difference between incremental and iterative is:

- A) Incremental adds functionality, Iterative refines existing functionality
- B) They are the same thing
- C) Iterative is faster
- D) Incremental is newer

Answer: A) Incremental adds functionality, Iterative refines existing functionality

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Prototyping Model

Q35. What is the primary purpose of a software prototype?

- A) To create the final production system
- B) To demonstrate concepts and try out design options
- C) To replace all documentation
- D) To eliminate the need for testing

Answer: B) To demonstrate concepts and try out design options

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Q36. A prototype can be used in which of the following processes?

- A) Requirements engineering only
- B) Design exploration only
- C) Testing only
- D) Requirements engineering, design exploration, and testing

Answer: D) Requirements engineering, design exploration, and testing

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Q37. Which of the following is NOT a benefit of prototyping?

- A) Improved system usability
- B) Closer match to users' real needs
- C) Eliminates the need for final testing
- D) Improved design quality

Answer: C) Eliminates the need for final testing

Q38. What is a throwaway prototype?

- A) A prototype that continues to be used in production
- B) A prototype discarded after development as it's not suitable for production
- C) A prototype that is never tested
- D) A prototype used for training only

Answer: B) A prototype discarded after development as it's not suitable for production

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Q39. Why should prototypes be discarded after development?

- A) To save storage space
- B) Because they may not meet non-functional requirements and lack proper documentation
- C) Because they are outdated
- D) To force developers to start over

Answer: B) Because they may not meet non-functional requirements and lack proper documentation

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Q40. The Prototype Model is characterized as:

- A) Top-down approach
- B) Bottom-up approach
- C) Middle-out approach
- D) No structured approach

Answer: B) Bottom-up approach

Q41. In the Prototype Development Process Model, what is the first phase?

- A) Define prototype functionality
- B) Establish prototype objectives
- C) Develop prototype
- D) Evaluate prototype

Answer: B) Establish prototype objectives

Q42. What is the main focus of the prototype when requirements are least understood?

- A) Non-functional requirements
- B) Business logic
- C) Areas not well understood
- D) User interface only

Answer: C) Areas not well understood

Q43. Prototyping is ideal for:

- A) Systems where requirements are fixed
- B) Simple batch processing systems
- C) Online systems with high user interaction
- D) Real-time embedded systems

Answer: C) Online systems with high user interaction

Q44. Which phase in the Prototype Development Process comes after "Develop prototype"?

- A) Define prototype functionality
- B) Establish prototype objectives
- C) Evaluate prototype
- D) Plan next phases

Answer: C) Evaluate prototype

Q45. A key advantage of prototyping is:

- A) Reduced development cost always
- B) Client involvement and early feedback
- C) No need for documentation
- D) Immediate deployment

Answer: B) Client involvement and early feedback

Q46. What is a disadvantage of the Prototype model?

- A) Too inflexible
- B) Customers may demand actual product after seeing prototype
- C) No user involvement possible
- D) Cannot be used for web applications

Answer: B) Customers may demand actual product after seeing prototype

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Spiral Model

Q47. Who developed the Spiral model?

- A) Winston Royce
- B) Barry Boehm
- C) Grady Booch
- D) Kent Beck

Answer: B) Barry Boehm

Q48. The Spiral model combines elements of:

- A) Waterfall and Big Bang
- B) Waterfall and Prototyping
- C) Agile and RUP
- D) Incremental and Big Bang

Answer: B) Waterfall and Prototyping

Q49. What does the spiral in Boehm's model represent?

- A) Project timeline
- B) Team structure
- C) Iterative cycles with different focus areas
- D) Budget allocation

Answer: C) Iterative cycles with different focus areas

Q50. In the Spiral model, what are the four main quadrants/sectors?

- A) Analysis, Design, Implementation, Testing
- B) Objective setting, Risk assessment, Development, Planning
- C) Requirements, Design, Code, Deploy
- D) Plan, Execute, Monitor, Control

Answer: B) Objective setting, Risk assessment, Development, Planning

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Q51. Which sector in the Spiral model addresses risk assessment?

- A) First quadrant (Objective setting)
- B) Second quadrant (Risk assessment and reduction)
- C) Third quadrant (Development and validation)
- D) Fourth quadrant (Planning)

Answer: B) Second quadrant (Risk assessment and reduction)

Q52. What is the main advantage of the Spiral model?

- A) Very fast development
- B) Minimal documentation required
- C) Risk-driven approach to development
- D) Simple for small projects

Answer: C) Risk-driven approach to development

Q53. A disadvantage of the Spiral model is:

- A) Cannot be used for large projects
- B) Risk analysis requires highly specific expertise
- C) No iteration
- D) Cannot handle changing requirements

Answer: B) Risk analysis requires highly specific expertise

Q54. When is the Spiral model most appropriate?

- A) For small projects with fixed requirements
- B) For medium to high risk projects with uncertain needs
- C) For projects with no budget constraints
- D) For real-time systems only

Answer: B) For medium to high risk projects with uncertain needs

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Q55. The Spiral model is rarely used in practice because:

- A) It's too simple
- B) It can be costly and requires risk analysis expertise
- C) It doesn't work for large projects
- D) It's incompatible with modern tools

Answer: B) It can be costly and requires risk analysis expertise

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Rapid Application Development (RAD)

Q56. RAD stands for:

- A) Rapid Automated Design
- B) Rapid Application Development
- C) Rapid Application Deployment
- D) Rapid Adaptive Development

Answer: B) Rapid Application Development

Q57. What is the primary characteristic of RAD?

- A) Extremely long development cycles
- B) Emphasizes very short development cycles
- C) Minimal client involvement
- D) No iteration

Answer: B) Emphasizes very short development cycles

Q58. How is high speed achieved in RAD?

- A) By having larger teams
- B) By using component-based project construction approach
- C) By skipping testing
- D) By reducing documentation

Answer: B) By using component-based project construction approach

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Q59. RAD can deliver a fully functional system within:

- A) 1-2 years
- B) 6-9 months
- C) 60-90 days
- D) 2-3 weeks

Answer: C) 60-90 days

Q60. Which of the following is a prerequisite for RAD success?

- A) Unclear requirements
- B) Well understood requirements with constrained scope
- C) Unlimited budget
- D) Minimal team size

Answer: B) Well understood requirements with constrained scope

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Q61. The first phase of RAD is:

- A) Data Modeling
- B) Business Modeling
- C) Process Modeling
- D) Application Generation

Answer: B) Business Modeling

Q62. In RAD, the Business Modeling phase answers which question?

- A) How does the system work?
- B) What are the business processes and information flow?
- C) How will we implement this?
- D) When will it be deployed?

Answer: B) What are the business processes and information flow?

Q63. What is the purpose of the Data Modeling phase in RAD?

- A) To write database code
- B) To define data objects and their relationships
- C) To test the database
- D) To deploy the database

Answer: B) To define data objects and their relationships

Q64. Process Modeling in RAD involves:

- A) Creating prototypes
- B) Writing code
- C) Transforming data objects to achieve information flow
- D) Testing procedures

Answer: C) Transforming data objects to achieve information flow

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Q65. What does the Application Generation phase in RAD focus on?

- A) Testing the application
- B) Deploying the application
- C) Reusing existing components and creating reusable components
- D) Documenting the application

Answer: C) Reusing existing components and creating reusable components

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Q66. The final phase of RAD is:

- A) Implementation
- B) Deployment
- C) Testing and Turnover
- D) Maintenance

Answer: C) Testing and Turnover

Q67. A critical drawback of RAD for large projects is:

- A) Cannot use modern tools
- B) Requires sufficient human resources to create right number of RAD teams
- C) Cannot handle databases
- D) Is incompatible with iterative development

Answer: B) Requires sufficient human resources to create right number of RAD teams

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Q68. RAD requires commitment from:

- A) Managers only
- B) Developers only
- C) Both developers and customers
- D) External consultants only

Answer: C) Both developers and customers

SECTION 3: SOFTWARE ENGINEERING FUNDAMENTALS

Requirements and Specifications

Q69. What is a requirement in software engineering?

- A) The number of developers needed
- B) A condition or capability needed by a user or system
- C) The project budget
- D) The deployment schedule

Answer: B) A condition or capability needed by a user or system

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Q70. Functional requirements define:

- A) How fast the system operates
- B) What the system should do
- C) How reliable the system is
- D) The cost of development

Answer: B) What the system should do

Q71. Non-functional requirements include:

- A) User interface design
- B) Business logic
- C) Performance, security, reliability
- D) Database structure

Answer: C) Performance, security, reliability

Software Process and Methodology

Q72. A software development process model is:

- A) The source code of the project
- B) A prescription for how to develop software
- C) The project budget
- D) The team's skill level

Answer: B) A prescription for how to develop software

Q73. Why is a well-defined software process important?

- A) To eliminate all errors
- B) To guide activities, tools, and practices
- C) To ensure quick deployment
- D) To reduce team size

Answer: B) To guide activities, tools, and practices

Software Engineering Motivation

Q74. The main goal of software engineering is:

- A) To write code quickly
- B) To produce high-quality software systematically
- C) To minimize team size
- D) To avoid documentation

Answer: B) To produce high-quality software systematically

Q75. Which is a key challenge in software engineering?

- A) Using only one programming language
- B) Managing complexity and change
- C) Avoiding all documentation
- D) Working in isolation

Answer: B) Managing complexity and change

SECTION 4: AGILE MODELS (from Lecture 5)

Agile Principles

Q76. The Agile philosophy emphasizes:

- A) Extensive documentation before coding
- B) Individuals and interactions over processes and tools
- C) Following a detailed plan rigidly
- D) Waterfall methodology

Answer: B) Individuals and interactions over processes and tools

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Q77. Which is a core principle of Agile?

- A) Complete documentation is essential before development
- B) Customer collaboration and regular feedback
- C) Minimal customer involvement
- D) No changes allowed after initial planning

Answer: B) Customer collaboration and regular feedback

Q78. Agile development prioritizes:

- A) Comprehensive planning
- B) Working software over comprehensive documentation
- C) Following contracts strictly
- D) Rigid process adherence

Answer: B) Working software over comprehensive documentation

Agile Methodologies

Q79. Scrum is characterized by:

- A) Waterfall phases
- B) Sprints of 2-4 weeks with daily stand-ups
- C) No iteration
- D) Extensive upfront design

Answer: B) Sprints of 2-4 weeks with daily stand-ups

Q80. What is a Sprint in Scrum?

- A) A quick code review
- B) A test phase
- C) A time-boxed iteration, usually 2-4 weeks
- D) A deployment cycle

Answer: C) A time-boxed iteration, usually 2-4 weeks

Q81. What is the role of the Product Owner in Scrum?

- A) Manage the development team
- B) Maintain and prioritize the product backlog
- C) Execute all technical tasks
- D) Deploy the software

Answer: B) Maintain and prioritize the product backlog

Q82. The Scrum Master in Agile is responsible for:

- A) Writing all the code
- B) Making all decisions
- C) Removing impediments and facilitating the Scrum process
- D) Testing the software

Answer: C) Removing impediments and facilitating the Scrum process

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Q83. What is a User Story in Agile?

- A) A detailed technical specification
- B) A narrative description of a feature from user perspective
- C) A bug report
- D) A deployment schedule

Answer: B) A narrative description of a feature from user perspective

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Q84. In Agile, acceptance criteria define:

- A) The team structure
- B) Conditions that must be met for a story to be complete
- C) The project timeline
- D) The budget allocation

Answer: B) Conditions that must be met for a story to be complete

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Q85. Kanban is an Agile method that:

- A) Uses sprints exclusively
- B) Visualizes workflow and limits work in progress
- C) Requires extensive documentation
- D) Uses waterfall phases

Answer: B) Visualizes workflow and limits work in progress

Q86. What is a key advantage of Agile over Waterfall?

- A) Less customer involvement needed
- B) Ability to accommodate changes and provide early feedback
- C) Easier to predict final costs
- D) Minimal documentation required

Answer: B) Ability to accommodate changes and provide early feedback

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Q87. In Extreme Programming (XP), pair programming means:

- A) Two teams working independently
- B) Two developers working on the same code simultaneously
- C) A developer paired with a tester
- D) Two projects running in parallel

Answer: B) Two developers working on the same code simultaneously

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Q88. Test-Driven Development (TDD) in Agile follows:

- A) Code first, then test
- B) Test first, then code
- C) Test and code simultaneously
- D) Test at the end only

Answer: B) Test first, then code

Q89. Continuous Integration in Agile means:

- A) Integrating code once at the end of project
- B) Frequently integrating code changes and running automated tests
- C) Never integrating until deployment
- D) Manual integration only

Answer: B) Frequently integrating code changes and running automated tests

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Q90. What is the Product Backlog in Scrum?

- A) A list of completed tasks
- B) A prioritized list of features and requirements
- C) The team's time cards
- D) The deployed features

Answer: B) A prioritized list of features and requirements

Q91. What is the Sprint Backlog?

- A) All work that will ever be done
- B) Work selected from Product Backlog for current sprint
- C) Completed work from past sprints
- D) Work rejected by the team

Answer: B) Work selected from Product Backlog for current sprint

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Q92. How often does the Development team meet in Scrum?

- A) Monthly planning sessions
- B) Daily stand-up meetings
- C) Quarterly reviews
- D) No regular meetings

Answer: B) Daily stand-up meetings

Q93. A Daily Stand-up in Scrum typically covers:

- A) What was done, what will be done, and any obstacles
- B) Budget discussions
- C) Hiring decisions
- D) Marketing strategies

Answer: A) What was done, what will be done, and any obstacles

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Q94. What is a Sprint Review in Scrum?

- A) A critical evaluation of the team's performance
- B) Demonstration of completed work to stakeholders
- C) A meeting to fire underperformers
- D) A code review session

Answer: B) Demonstration of completed work to stakeholders

Q95. Sprint Retrospective is conducted to:

- A) Blame team members for issues
- B) Review completed features only
- C) Identify improvements for the next sprint
- D) Plan new features

Answer: C) Identify improvements for the next sprint

Q96. In Agile, what is the Definition of Done?

- A) When the project is deployed
- B) Criteria that define when a task/story is complete
- C) The end of the project
- D) When customers stop requesting changes

Answer: B) Criteria that define when a task/story is complete

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Q97. Agile is most suitable for:

- A) Projects with completely fixed requirements
- B) Projects requiring frequent change and customer feedback
- C) Large, long-term waterfall projects
- D) Systems with no user interaction

Answer: B) Projects requiring frequent change and customer feedback

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Q98. What is a User Story Point in Agile estimation?

- A) The actual number of days to complete
- B) Relative measure of complexity/effort
- C) The cost per feature
- D) The number of bugs

Answer: B) Relative measure of complexity/effort

Q99. Which Agile ceremony/meeting is most important for team synchronization?

- A) Monthly all-hands meetings
- B) Quarterly business reviews
- C) Daily stand-up meetings
- D) Annual planning sessions

Answer: C) Daily stand-up meetings

Q100. In Agile, who is responsible for determining if a feature is complete?

- A) The Project Manager
- B) The Scrum Master only
- C) The Development Team and Product Owner
- D) The CEO

Answer: C) The Development Team and Product Owner

SECTION 5: COMPARISON QUESTIONS

Q101. Waterfall vs. Agile - Which requires extensive upfront planning?

- A) Agile
- B) Waterfall
- C) Both equally
- D) Neither

Answer: B) Waterfall

Q102. Which model is best when requirements are unclear?

- A) Waterfall
- B) RAD
- C) Iterative/Agile
- D) Big Bang

Answer: C) Iterative/Agile

Q103. Which model emphasizes risk analysis?

- A) Waterfall
- B) Spiral
- C) Prototyping
- D) RAD

Answer: B) Spiral

Q104. Which model is most appropriate for systems where user interface design is critical?

- A) Waterfall
- B) Prototyping
- C) RAD
- D) Big Bang

Answer: B) Prototyping

Q105. Comparing Incremental vs. Iterative - Which one refines existing functionality?

- A) Incremental
- B) Iterative
- C) Both equally
- D) Neither

Answer: B) Iterative

Q106. Which traditional model is most similar to Agile?

- A) Waterfall
- B) Spiral
- C) Incremental/Iterative
- D) RAD

Answer: C) Incremental/Iterative

Q107. Which model works best for very tight time constraints?

- A) Waterfall
- B) Prototyping
- C) RAD
- D) Spiral

Answer: C) RAD

Q108. In terms of customer involvement, which model requires the most?

- A) Waterfall
- B) Prototyping
- C) Agile
- D) Big Bang

Answer: C) Agile

SECTION 6: APPLICATION & SCENARIO-BASED

Q109. A client has vague requirements for a web application. Which development model would you recommend?

- A) Waterfall
- B) Iterative/Agile with prototyping
- C) Big Bang model
- D) Linear sequential model

Answer: B) Iterative/Agile with prototyping

Q110. A critical banking system with high security and reliability needs. Which model fits best?

- A) RAD
- B) Agile with short sprints
- C) Spiral (risk-driven) or Waterfall with heavy testing
- D) Prototyping only

Answer: C) Spiral (risk-driven) or Waterfall with heavy testing

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Q111. You need to deliver a simple CRM system in 90 days with clear requirements. Which model?

- A) Waterfall
- B) RAD
- C) Spiral
- D) Prototyping

Answer: B) RAD

Q112. A new emerging technology with uncertain requirements requires exploration. Which model?

- A) Waterfall with strict timelines
- B) RAD with fixed scope
- C) Iterative/Spiral with frequent risk assessment
- D) Big Bang approach

Answer: C) Iterative/Spiral with frequent risk assessment

Q113. Your team knows the exact requirements and domain. Best approach?

- A) Extended prototyping phase
- B) Waterfall or RAD
- C) Spiral model
- D) Iterative without planning

Answer: B) Waterfall or RAD

Q114. A user provides a user story: "As a student, I want to register for courses, so I can enroll in my preferred classes." What is the "what" (goal)?

- A) Student
- B) Register for courses
- C) Enroll in preferred classes
- D) Course registration system

Answer: B) Register for courses

Q115. In the user story example above, what is the "who"?

- A) The system
- B) A student
- C) The administrator
- D) The institution

Answer: B) A student

Q116. For a user story "register for courses," which of these would be good acceptance criteria?

- A) The system should be fast
- B) The student can select courses from an available list and confirm registration
- C) The database should be efficient
- D) Users should be happy

Answer: B) The student can select courses from an available list and confirm registration

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Q117. A complex story about "online banking system" should be:

- A) Implemented as a single user story
- B) Split into smaller stories following vertical slicing
- C) Not split at all
- D) Implemented immediately

Answer: B) Split into smaller stories following vertical slicing

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Q118. Which component is missing from a user story that lacks acceptance criteria?

- A) Conversation
- B) Confirmation (how do we know it's done?)
- C) Description
- D) Priority

Answer: B) Confirmation (how do we know it's done?)

SECTION 7: CONCEPT MATCHING

Q119. Match: "Process that discovers what and how you need as you go"

- A) Waterfall
- B) Incremental
- C) Iterative
- D) RAD

Answer: C) Iterative

Q120. Match: "Process that slices product into chunks and builds separately"

- A) Waterfall
- B) Incremental
- C) Iterative
- D) Spiral

Answer: B) Incremental

Q121. Match: "Model that emphasizes component-based rapid development"

- A) Waterfall
- B) Prototype
- C) RAD
- D) Spiral

Answer: C) RAD

Q122. Match: "Model that explicitly assesses and resolves risks"

- A) Waterfall
- B) Prototype
- C) RAD
- D) Spiral

Answer: D) Spiral

Q123. Match: "Process used to explore uncertain requirements"

- A) RAD
- B) Waterfall
- C) Prototype
- D) Incremental

Answer: C) Prototype

Q124. Match: "Agile framework emphasizing time-boxed iterations"

- A) Kanban
- B) Scrum
- C) XP
- D) SAFe

Answer: B) Scrum

Q125. Match: "Agile method that limits work in progress"

- A) Scrum
- B) Kanban
- C) XP
- D) Waterfall

Answer: B) Kanban

SECTION 8: TRUE/FALSE QUESTIONS

Q126. User stories must be written by developers.

- A) True
- B) False

Answer: B) False (They should be written from stakeholder perspective)

Q127. The Waterfall model allows overlapping of phases.

- A) True
- B) False

Answer: B) False (Each phase must complete before next begins)

Q128. Prototypes are always meant to be used in production.

- A) True
- B) False

Answer: B) False (Prototypes are typically throwaway)

Q129. The Spiral model is the most commonly used model in practice.

- A) True
- B) False

Answer: B) False (It's rarely used due to complexity and cost)

Q130. In Agile, customer involvement is important.

- A) True
- B) False

Answer: A) True

Q131. Acceptance criteria define the conditions for story completion.

- A) True
- B) False

Answer: A) True

Q132. Iterative development requires no feedback from stakeholders.

- A) True
- B) False

Answer: B) False (Feedback is essential)

Q133. RAD can deliver systems in 60-90 days when requirements are well-defined.

- A) True
- B) False

Answer: A) True

Q134. User stories are detailed technical specifications.

- A) True
- B) False

Answer: B) False (They are brief reminders for conversation)

Q135. The Conversation component of a user story captures how we know it's done.

- A) True
- B) False

Answer: B) False (That's the Confirmation component)

SECTION 9: FILL IN THE BLANKS

Q136. User stories seek to combine _____ and verbal communication.

- A) Written documentation
- B) Code
- C) Visual design
- D) Database schema

Answer: A) Written documentation

Q137. In incremental development, _____ must be known upfront.

- A) All requirements
- B) All resources
- C) All risks
- D) All team members

Answer: A) All requirements

Q138. Iterative development uses a process of _____.

- A) Fixed planning
- B) Successive approximation
- C) Linear progression
- D) Parallel processing

Answer: B) Successive approximation

Q139. A prototype should focus on areas _____ understood.

- A) Well
- B) Not well
- C) Partially
- D) Completely

Answer: B) Not well

Q140. The Spiral model has _____ main sectors/quadrants.

- A) Two
- B) Three
- C) Four
- D) Five

Answer: C) Four

SECTION 10: PRIORITY & IMPORTANCE

Q141. According to INVEST, which property is most important?

- A) Size
- B) Estimatable
- C) Testable
- D) All are equally important

Answer: D) All are equally important (All INVEST properties are necessary)

Q142. In a user story, which part is most critical for developers?

- A) The user role
- B) The goal/what needs to be done
- C) The reason/why
- D) The status

Answer: B) The goal/what needs to be done

Q143. Which aspect of Agile is most critical for project success?

- A) Comprehensive documentation
- B) Regular customer feedback and collaboration
- C) Detailed upfront planning
- D) Large team size

Answer: B) Regular customer feedback and collaboration

ANSWER KEY SUMMARY

Total MCQs Created: 143

Distribution by Topic:

- User Stories: Q1-Q25 (25 questions)
- Traditional Models: Q26-Q68 (43 questions)
- Software Engineering Fundamentals: Q69-Q75 (7 questions)
- Agile Models: Q76-Q100 (25 questions)
- Comparison Questions: Q101-Q108 (8 questions)
- Application & Scenarios: Q109-Q118 (10 questions)
- Concept Matching: Q119-Q125 (7 questions)
- True/False: Q126-Q135 (10 questions)
- Fill in Blanks: Q136-Q140 (5 questions)
- Priority & Importance: Q141-Q143 (3 questions)

HOW TO USE THESE MCQs

1. **Self-Assessment**: Use these MCQs to evaluate your understanding
2. **Study Guide**: Refer to lecture material for explanations of answers
3. **Exam Preparation**: Practice repeatedly to build confidence
4. **Topic Review**: Focus on specific sections based on weak areas
5. **Concept Connection**: Compare answers to understand relationships between concepts

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DIFFICULTY LEVELS

Easy (Foundational): Q1, Q2, Q26, Q56, Q76, Q79

Medium (Applied): Q14-Q25, Q50-Q60, Q86-Q95

Hard (Analysis): Q101-Q118, Q119-Q143

KEY TAKEAWAYS FOR EXAM PREPARATION

User Stories Remember:

- Written from stakeholder perspective, not developer
- Have 3 parts: Description, Conversation, Confirmation
- Follow INVEST criteria
- Vertical slicing approach for splitting
- Acceptance criteria define done

Traditional Models Remember:

- **Waterfall**: Sequential, all upfront, difficult to change
- **Incremental**: Add functionality in chunks
- **Iterative**: Refine through feedback loops
- **Prototype**: Explore uncertain areas
- **Spiral**: Risk-driven, iterative, expensive
- **RAD**: Fast delivery (60-90 days), well-defined requirements

Agile Models Remember:

- Customer collaboration over contracts
- Working software over documentation
- Responding to change over plans
- Scrum uses sprints, daily standups, product backlog
- User stories are core unit of work
- Continuous feedback and integration
